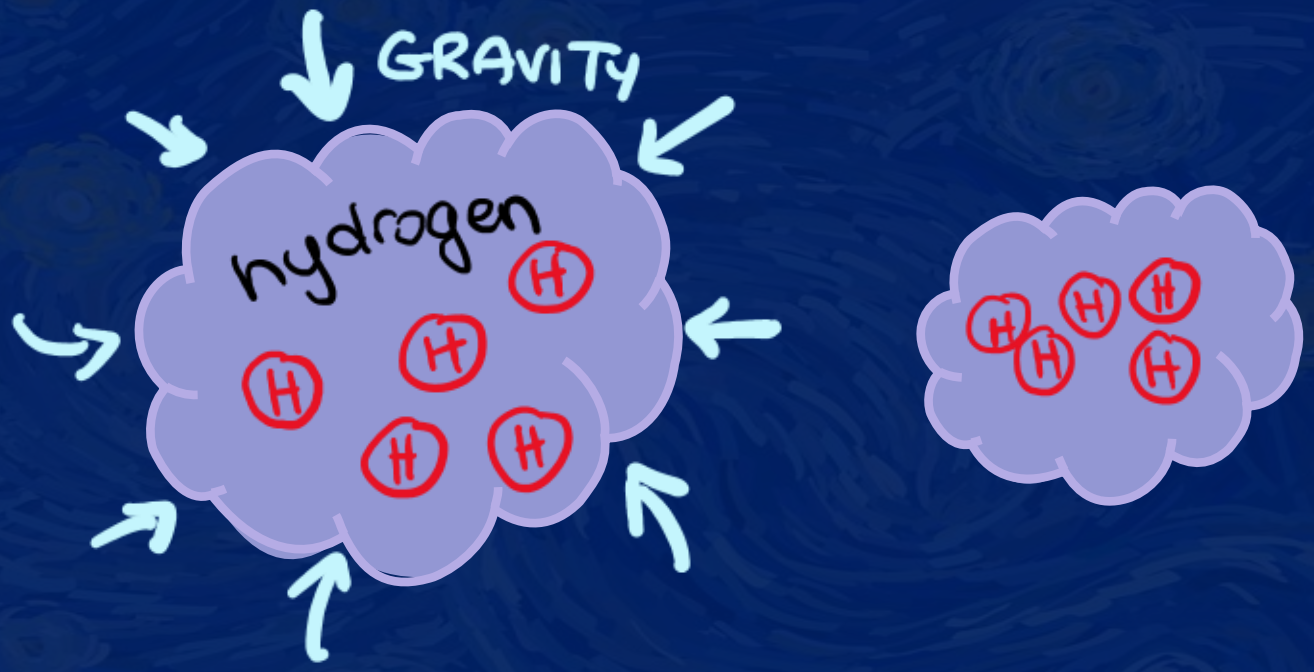
THE LIFE AND DEATHS OF STARS

YILDA BOUKHTOUCHEN
PHD CANDIDATE

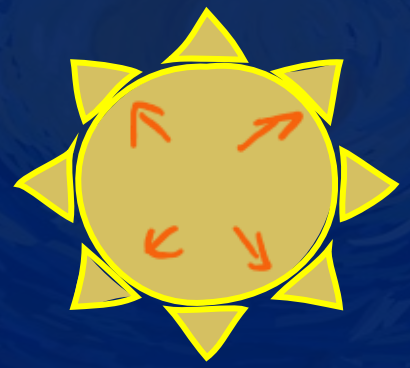
@

ASTRO ON TAP
FEB. 10, 2026

THE BIRTH OF A STAR



"pressure cooker"
allows H → He fusion

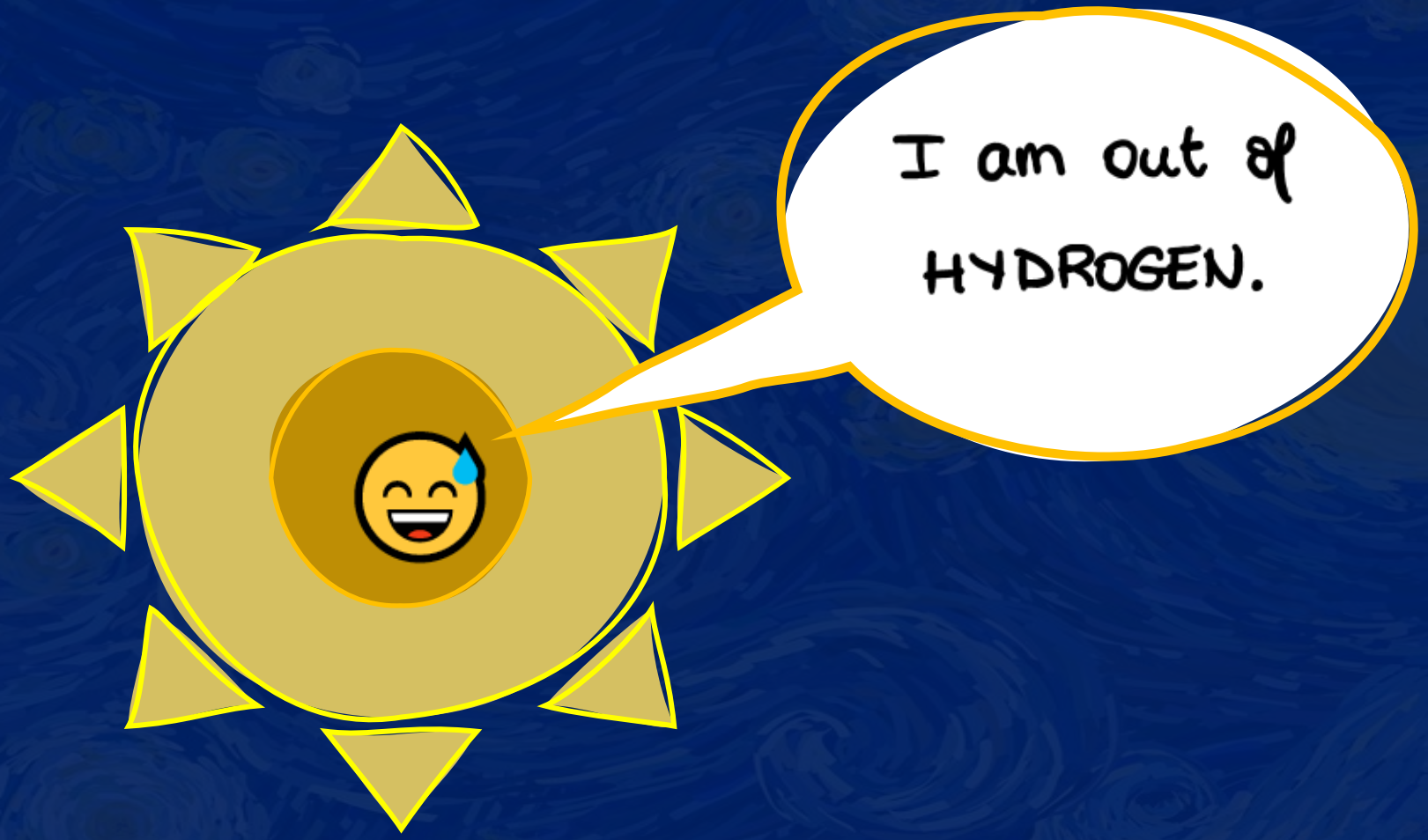


fusion releases
ENERGY
outward pressure

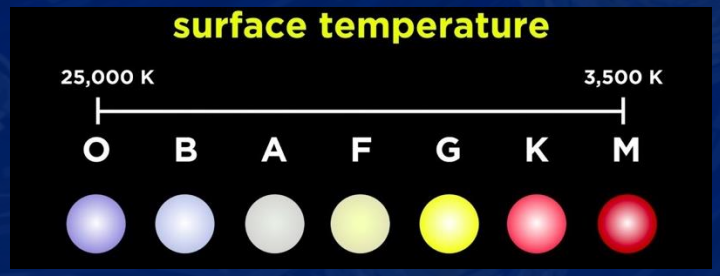
A STAR'S LIFE IS ALL ABOUT

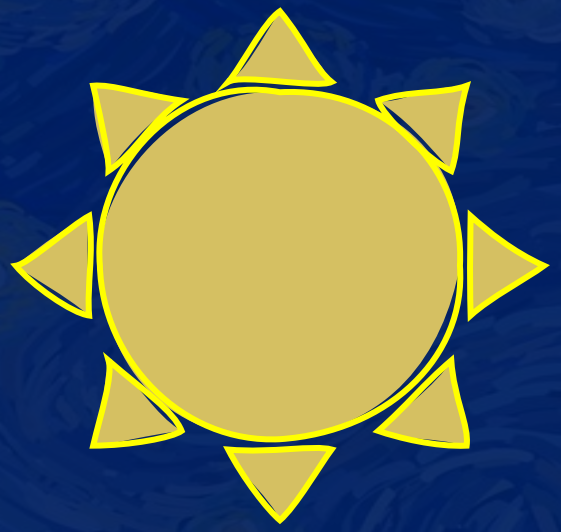
BALANCE!

UH OH.



QUIZ: WHO RUNS OUT FIRST?

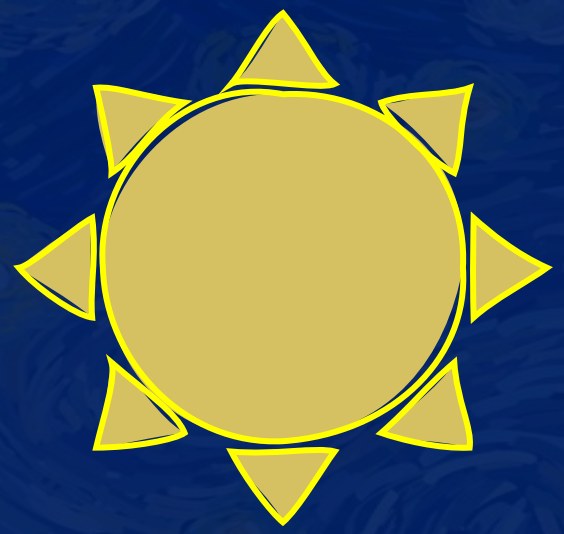




GEMMA THE "G" STAR
M = 1 sun

OBELIX THE "O" STAR
M = 10 suns

QUIZ: WHO RUNS OUT FIRST?



GEMMA THE "G" STAR

M = 1 sun

~ 10 billion years

more pressure

↓

higher temp

↓

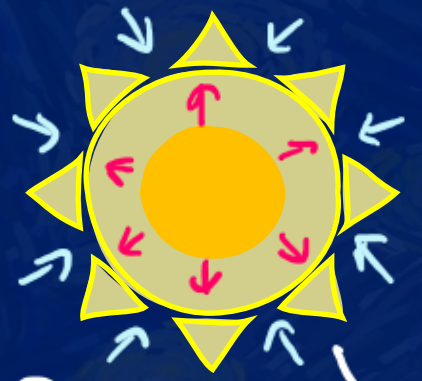
faster fusion

OBELIX THE "O" STAR

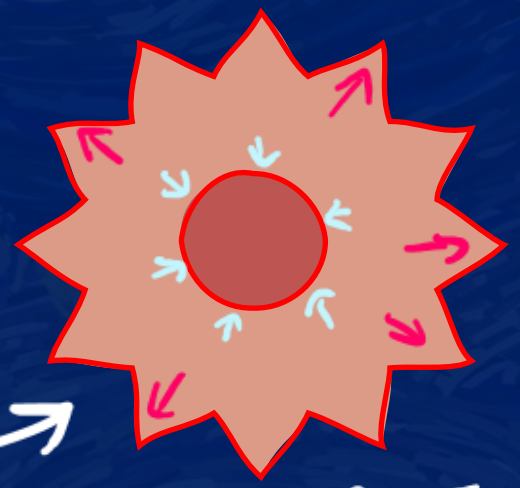
M = 10 suns

~ 100 million years

THE STAR IS TURNING RED (GIANT)



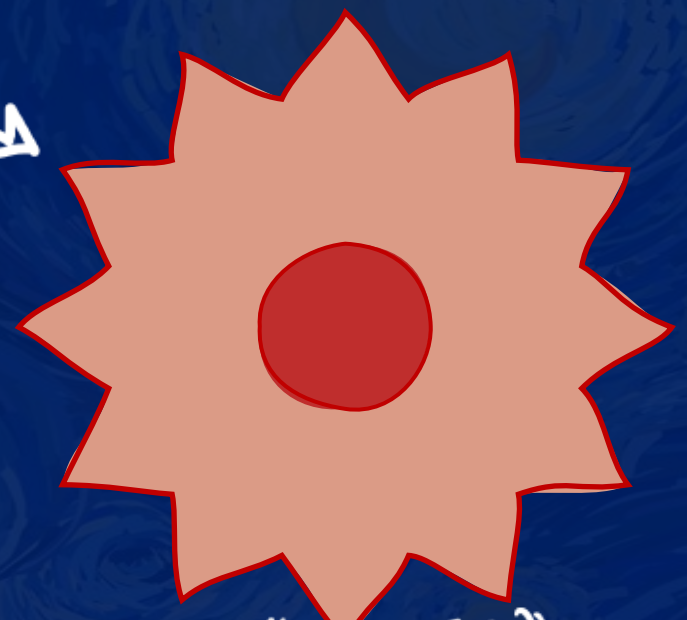
SUN-LIKE STAR



RED GIANT

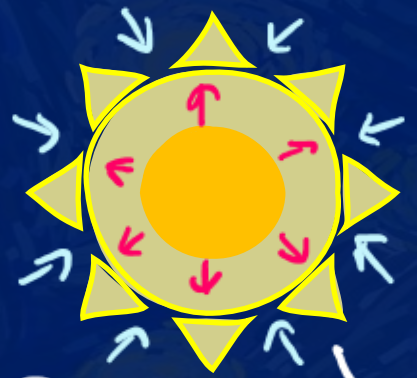
- core contracts til hot enough to fuse other elements
  3 helium → 1 carbon

MASSIVE STAR

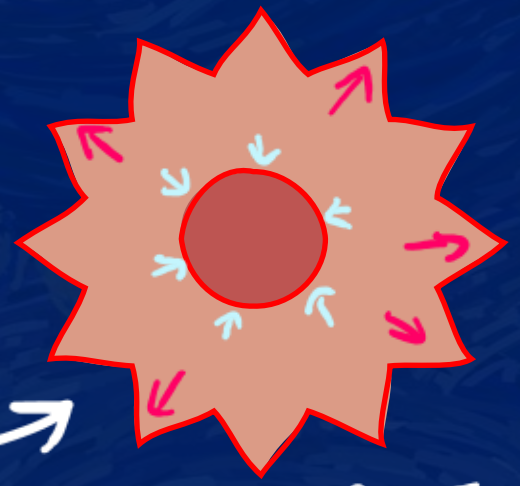


RED "SUPER" GIANT

THE STAR IS TURNING RED (GIANT)



SUN-LIKE STAR

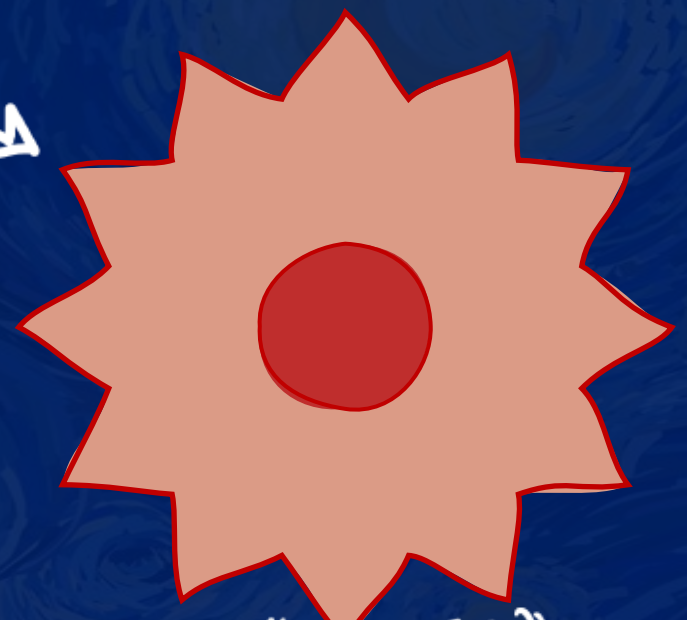


RED GIANT

- core contracts til hot enough to fuse other elements
  3 helium → 1 carbon

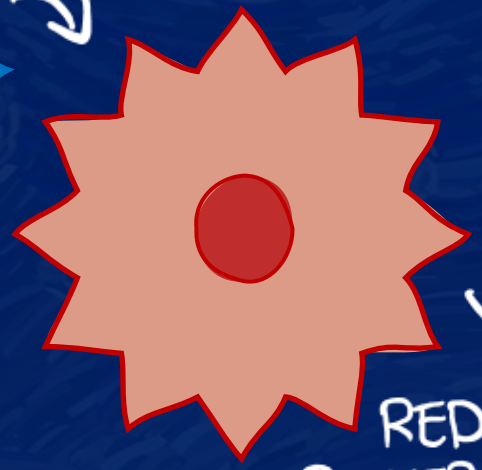
- hydrogen in outer layers fuses to helium
  ↳ heat outer layers
  ↳ expand & become "puffy"

MASSIVE STAR

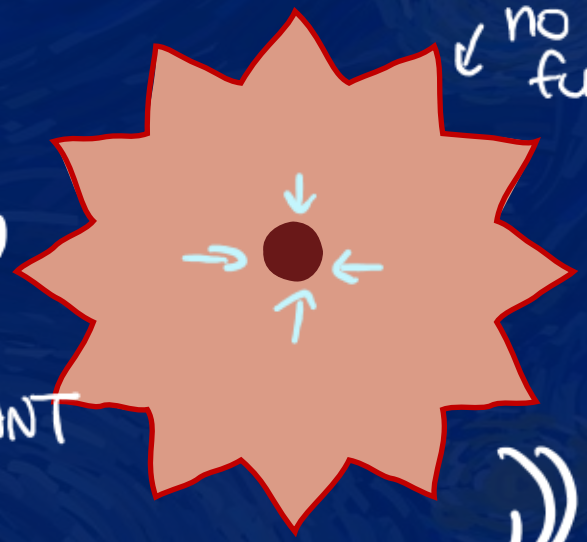


RED "SUPER" GIANT

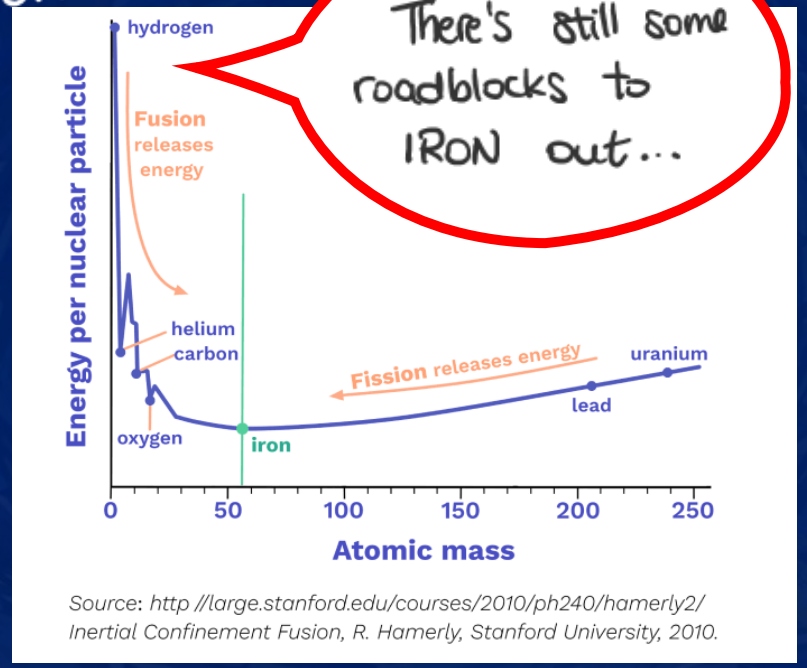
WHEN'S THE SUPERNOVA?



RED SUPERGIANT

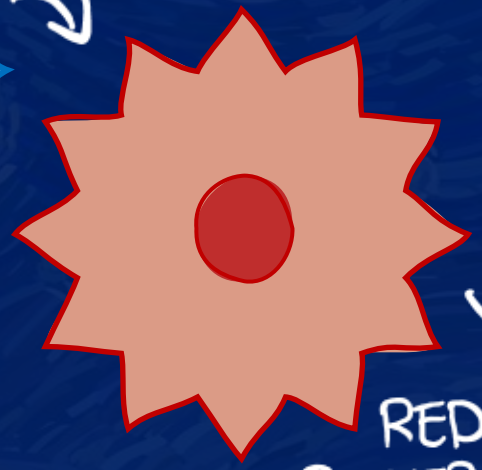


no more fusion...

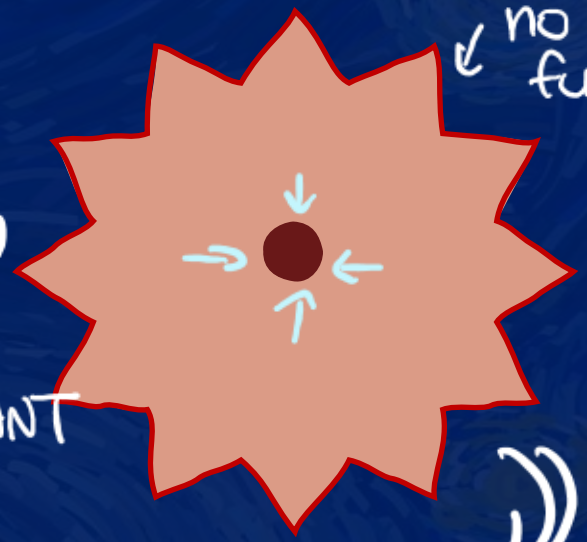


Source: http //large.stanford.edu/courses/2010/ph240/hamerly2/
Inertial Confinement Fusion, R. Hamerly, Stanford University, 2010.

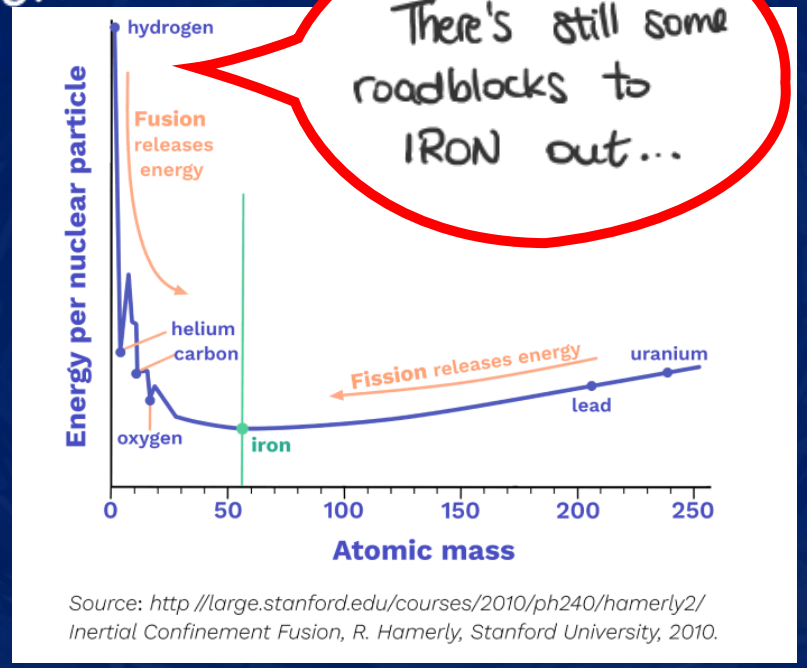
WHEN'S THE SUPERNOVA?



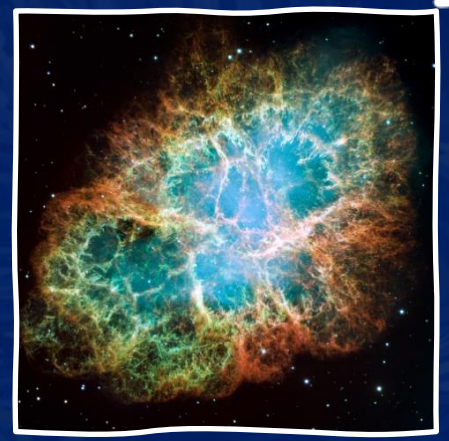
RED SUPERGIANT



no more fusion...



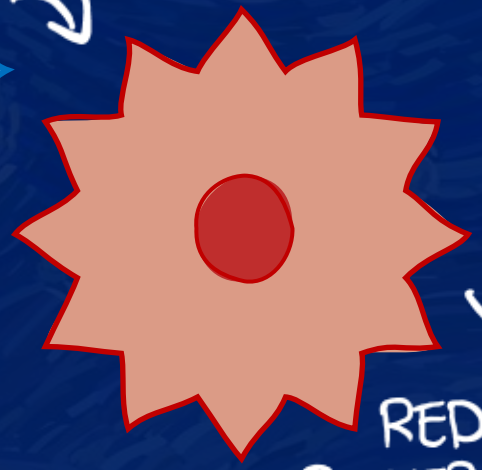
Source: http //large.stanford.edu/courses/2010/ph240/hamerly2/
Inertial Confinement Fusion, R. Hamerly, Stanford University, 2010.



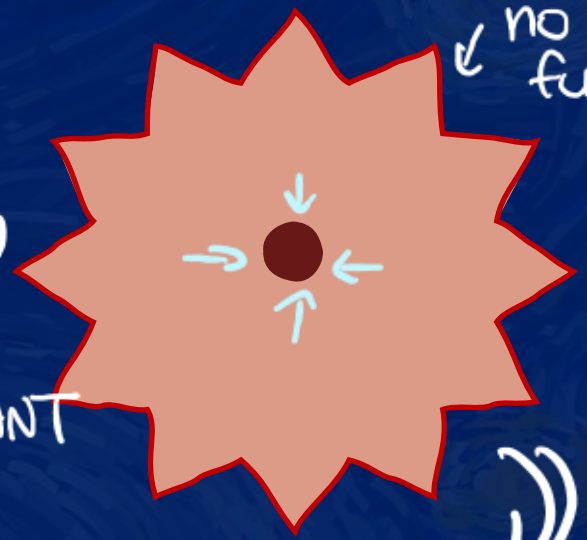
"core collapse" SUPERNOVA

"superior" + "new" (star)

WHEN'S THE SUPERNOVA?

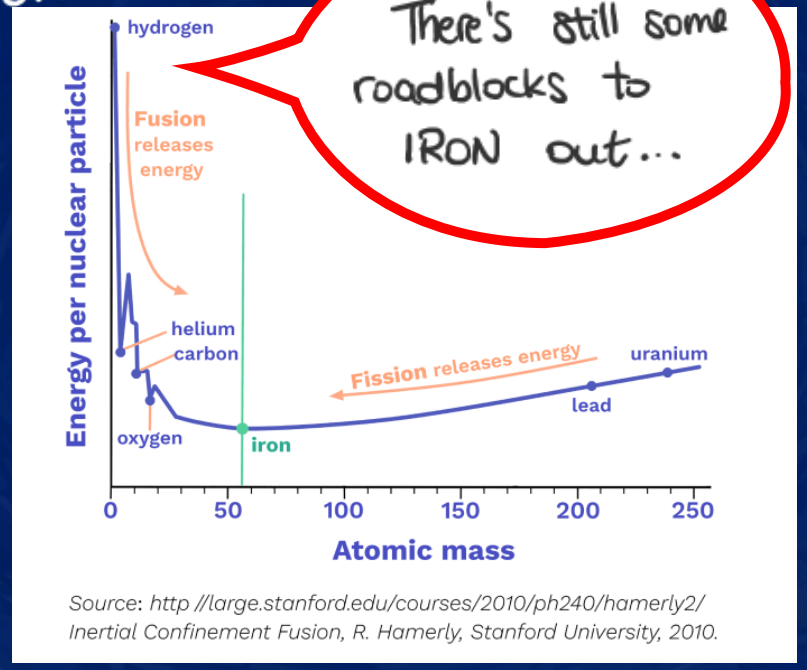


RED SUPERGIANT



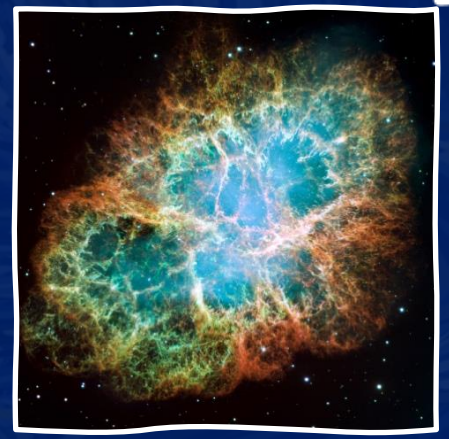
no more fusion...

There's still some roadblocks to IRON out...



Source: http //large.stanford.edu/courses/2010/ph240/hamerly2/ Inertial Confinement Fusion, R. Hamerly, Stanford University, 2010.

neutron star

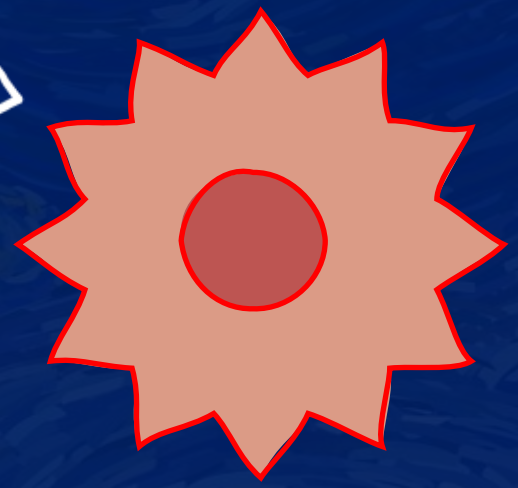


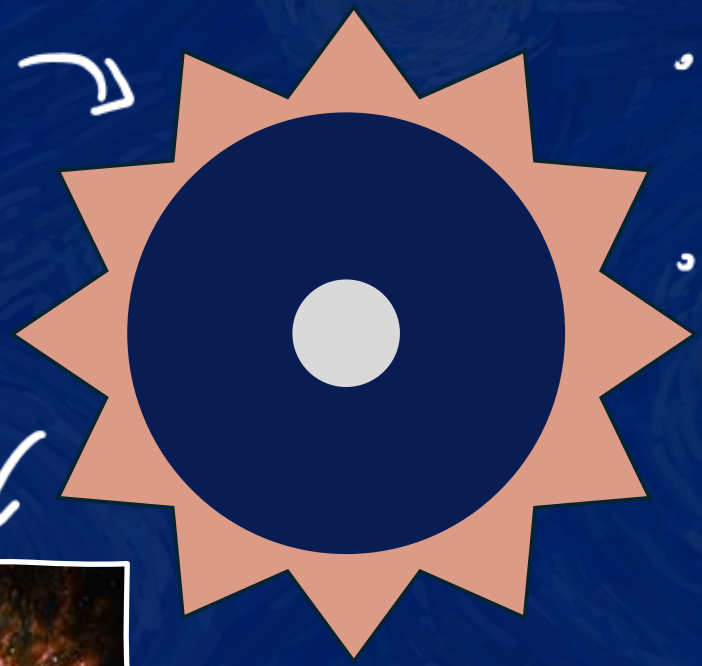
black hole

"core collapse" SUPERNOVA

"superior" + "new" (star)

WHAT ABOUT OUR SUN?





- outer layers are very unstable
- core is not massive enough to hold on!

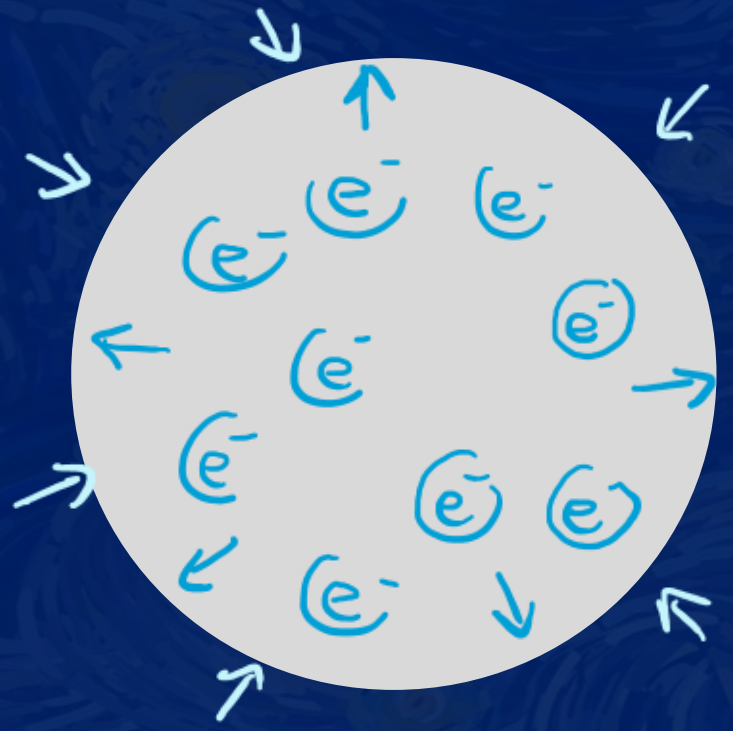
WHITE DWARF

dense remnant

"PLANETARY NEBULA"

outer gas is ejected

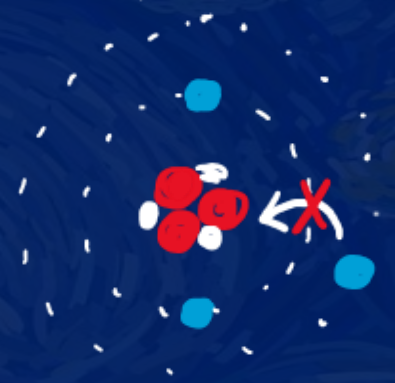
A NEW BALANCE IS FORMED!



WHITE DWARF

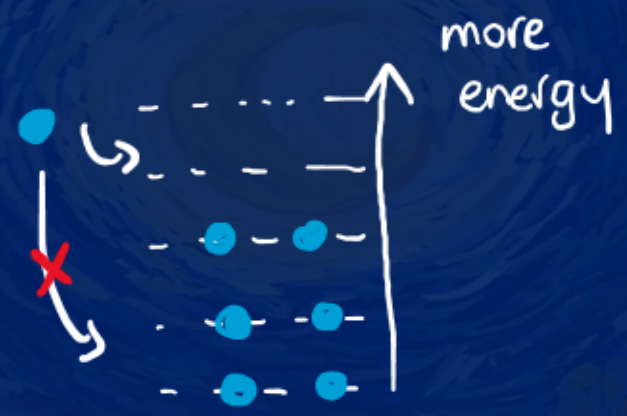
ELECTRON DEGENERACY PRESSURE

- Pauli exclusion: electrons like to be unique!



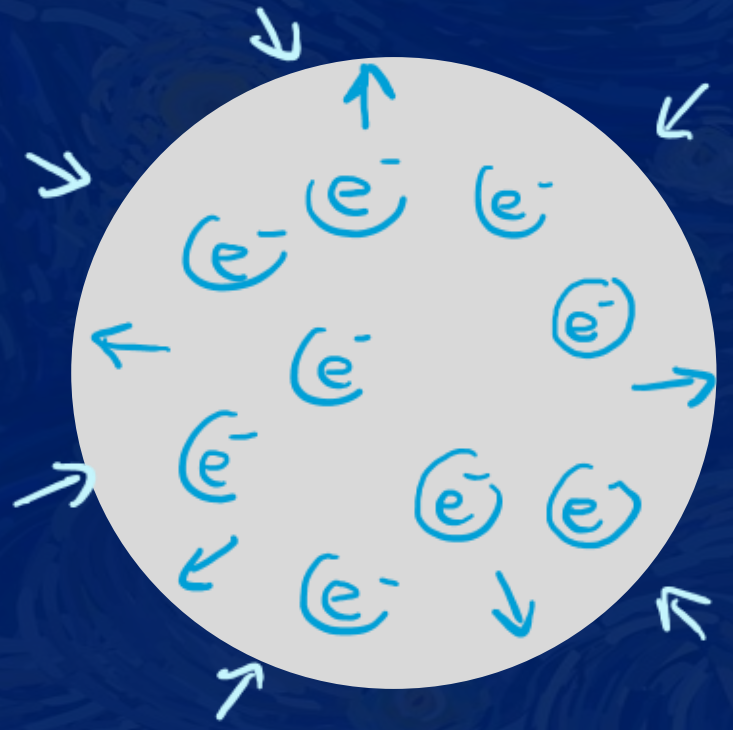
lithium atom

OR



momentum states

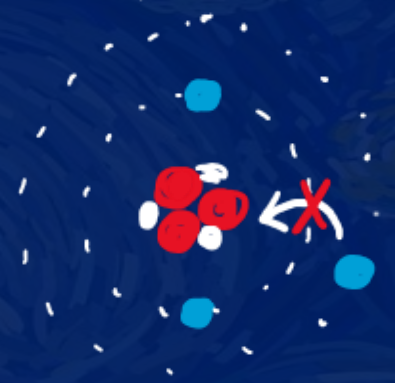
A NEW BALANCE IS FORMED!



WHITE DWARF

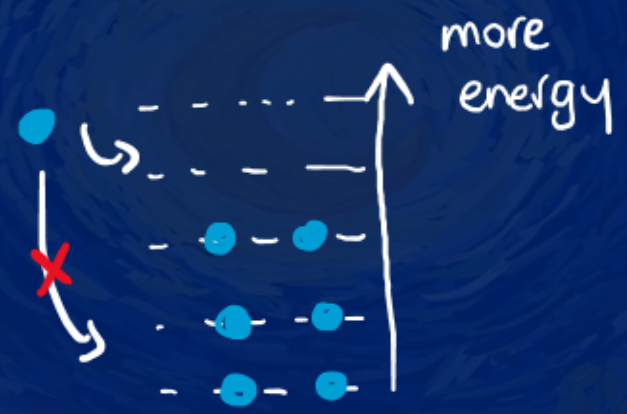
ELECTRON DEGENERACY PRESSURE

- Pauli exclusion: electrons like to be unique!



lithium atom

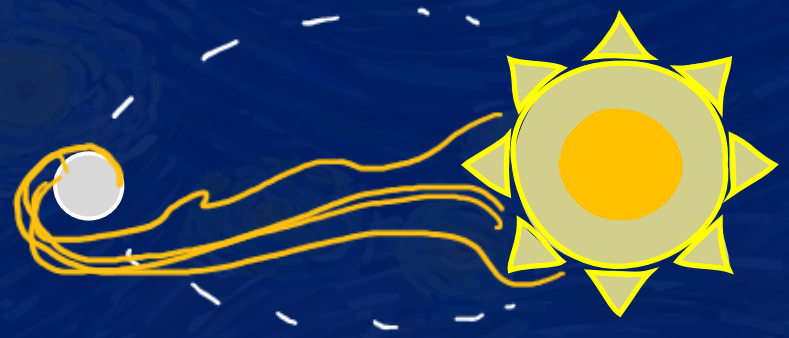
or



momentum states

- new outward pressure to counter gravity!

BUT THE PEACE IS NOT FOREVER



STELLAR BINARY SYSTEM

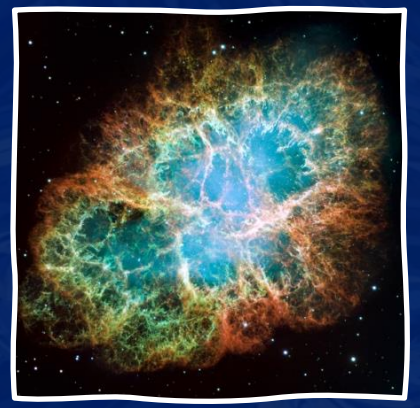
ACCRETION increases mass

"Chandrasekhar" limit

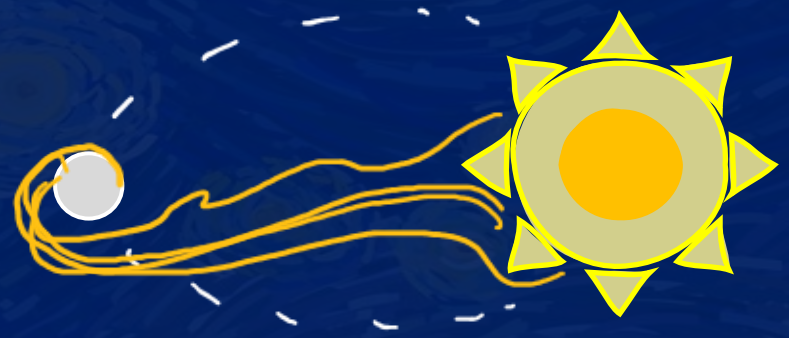
Star collapses

Carbon fusion

TYPE IA SUPERNOVA



BUT THE PEACE IS NOT FOREVER



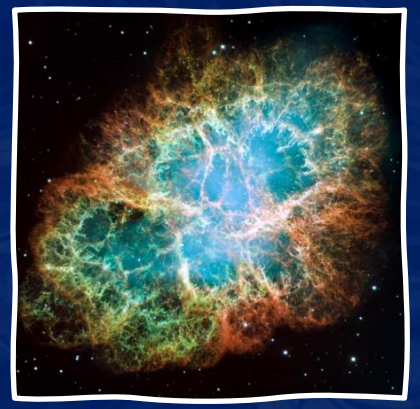
STELLAR BINARY SYSTEM

ACCRETION increases mass

"Chandrasekhar" limit

star collapses

Carbon fusion

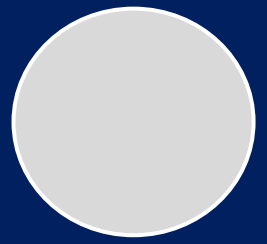


TYPE IA SUPERNOVA

COLLISION

energy to continue fusion

BUT WHAT IF… ?

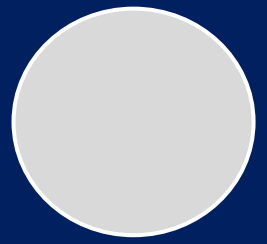


WENDY THE W.D.

- ☑ no neighbours to accrete
- ☑ no enemy to collide with

PEACE ?

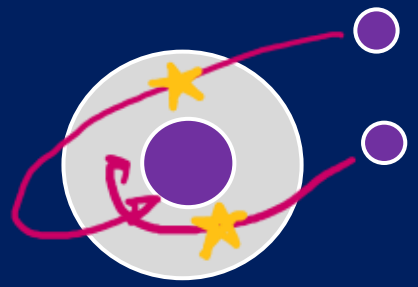
BUT WHAT IF… DARK MATTER?



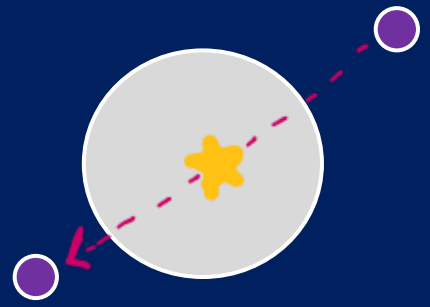
WENDY THE W.D.

- ☑ no neighbours to accrete
- ☑ no enemy to collide with

~~PEACE?~~ MAYBE NOT!



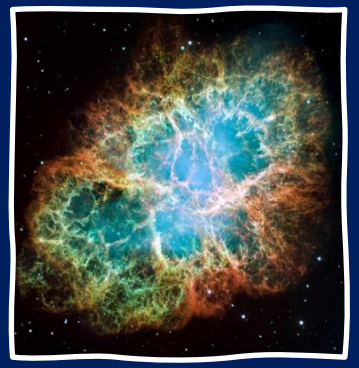
CAPTURE & COLLAPSE

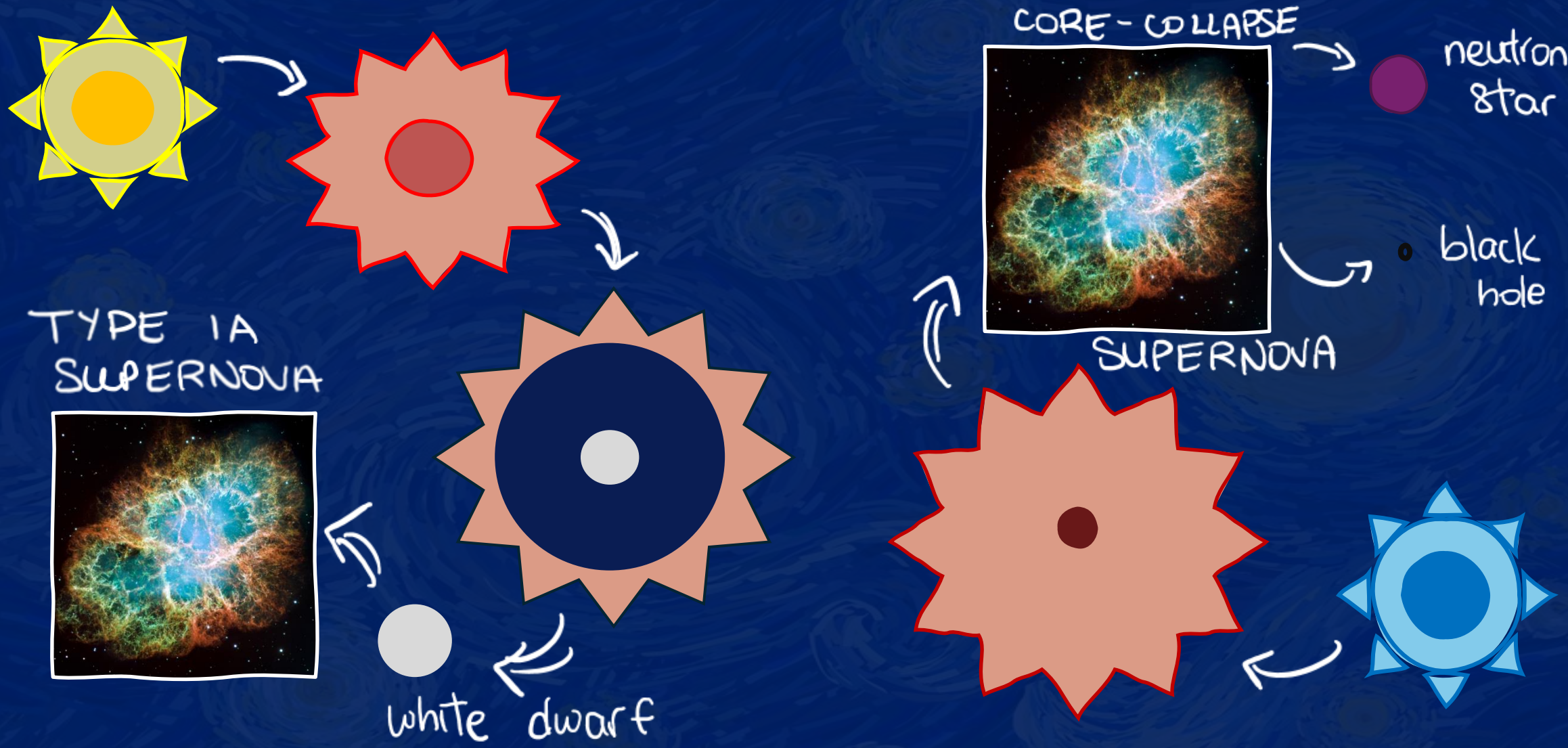


DEPOSIT ENERGY

BONUS

TYPE IA SUPERNOVA



CHEAT

THE LIFE AND DEATHS OF STARS